

Tab 1

RB Evidence Pack v1.0

1. Founder Identity

Name: Rishabh Banerjee

Also Known As: Rishi / Rish / Hidden Bong

Role: Founder & AI Systems Architect

Affiliation: R_B Tradelink Pvt. Ltd.

Jurisdiction: India

2. Authorship Declaration

This document serves as a formal, timestamped record of authorship, development, and intellectual direction for a set of interconnected systems and frameworks developed by Rishabh Banerjee.

The purpose of this record is to:

- establish a verifiable timeline of work
 - define clear authorship boundaries
 - provide traceable evolution of systems
 - enable external verification of integrity via cryptographic hashing
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3. Scope of Authorship

The following systems and conceptual frameworks are included within this Evidence Pack:

Core System Artifacts

Z.DecisionKernel v0.1.0-alpha (2026)

A governance-first decision auditing mechanism with rule-based evaluation, structured outputs, and auditability constraints.

Supply Chain Predictive & Correction Matrix (2024)

A system-level approach to integrating AI-based predictive modeling with operational optimization, risk reduction, and customer adaptation in supply chain environments.

System-Level Architecture (Referenced)

RGPT++

A trust-first AI framework focused on privacy, governance, modular runtime logic, and system-level intelligence.

.RB++ Vault (Conceptual Root)

A proposed identity and provenance system designed for append-only records, authorship anchoring, and trust-based system memory.

Goa Motoroid

A mobility and operational system applying trust, compliance, and structured governance to real-world rental and transportation ecosystems.

4. Timeline Reference

24 September 2024

- Supply Chain Predictive Model (LinkedIn publication)

2025–2026

- Iterative development of governance-based systems, trust architecture, and decision modeling

19 April 2026

- Z.DecisionKernel v0.1.0-alpha

5. Nature of Claim

This document asserts:

- authorship of the specific implementations, structures, and system relationships described herein
- continuity of development across the defined timeline

This document does NOT assert:

- exclusive ownership of general ideas within AI, supply chain, or governance domains
- claims of duplication or misuse by external entities without verifiable evidence

6. Integrity & Verification

This document is finalized as:

RB Evidence Pack v1.0

A cryptographic hash (SHA-256) will be generated for this exact version to establish:

- immutability
- timestamped existence
- integrity of content

Any modification to this document will result in a different hash value and will constitute a new version.

7. Statement of Intent

This Evidence Pack is created to:

- transition conceptual and internal work into verifiable external existence
- establish a non-ambiguous authorship record
- serve as a foundation for future development, validation, and collaboration

18:00 ↗



 ****R.B_Tradelink_Analytical Analysis:
Open for Endorsements on Business
Development Models & Strategies****

 ****Looking to Partner**:**
Seeking endorsements and collaboration opportunities to further develop cutting-edge ****predictive models**** for supply chain management, leveraging ****AI, machine learning, and data analytics****.

Key Strategic Areas:

- 1. **Business Model Innovation**:** Expanding offerings with dynamic predictive models that anticipate market shifts.
- 2. **Partnerships**:** Interested in teaming up with ****tech innovators****, logistics firms, and ****data-driven platforms**** to enhance operational efficiency.
- 3. **Market Penetration**:** Exploring new markets through data-driven insights and AI-optimized solutions for global supply

- **Optimized Operations:** Use AI/ML for precise demand forecasting and inventory management.
- **Reduced Risks:** Early detection of disruptions using real-time data analytics.
- **Improved Customer Satisfaction:** Meeting demand efficiently, reducing delays.

Future Trends:

- **Autonomous supply chains with AI.**
- **Blockchain for transparent forecasting.**
- **Digital twins for dynamic scenario planning.**

Let's collaborate to shape the future of supply chains!

****Prepared by**:-**

Rishabh Banerjee

| **Date: 24/09/2024 |**

****Co-Powered by**:-**

ChatGPT_4.0

2025–2026

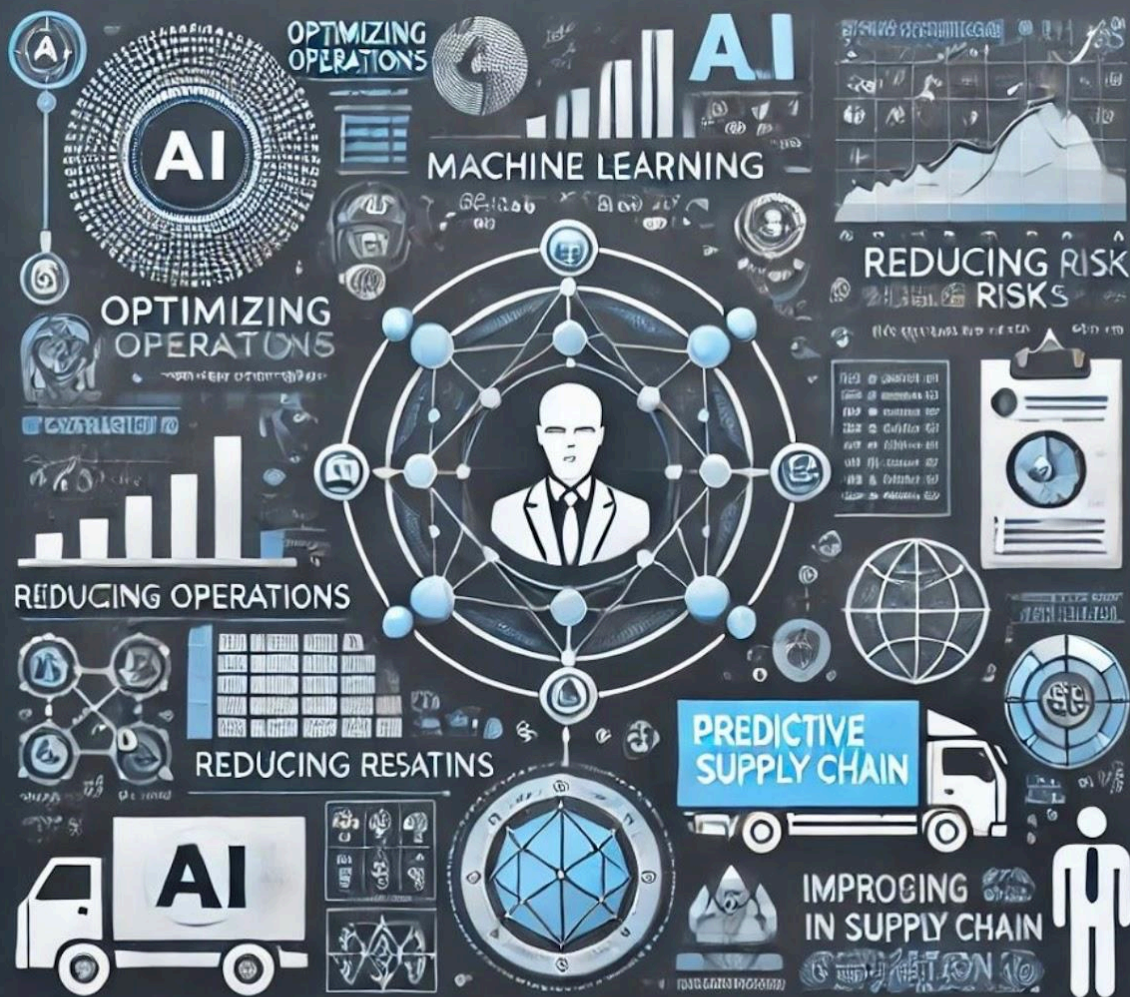
– System architecture developments

Note: AI tools were used as assistive instruments.

Conceptual authorship, system design, and structuring are attributed to Rishabh Banerjee.

R.B – PREDICTIVE MODELS IN SUPPLY CHAIN

PREDICTING DEMANDS - IMPROVING CUSTOMER SATISFACTION



Prepared by: R. Tradelinke — Rishabh Banerjee — Rishabh Banerjee
Date: 24/09/2024 — Date: 24/9/2024 — Improving Customer Satisfaction

19 April 2026

– Z.DecisionKernel v0.1.0- alpha

•Z.DecisionKernel— v0.1.0-alpha Specification

Version: 0.1.0-alpha Status: Frozen for reference use; not yet validated against

empirical runs Release Date: 2026-04-19 Primary Author: Rishabh Banerjee

Collaborators: ChatGPT, Perplexity, Gemini, Claude (Anthropic) Scope: Three-option

career / strategic-pivot decisions Intent: Governance-first decision auditing. Converts

ranked options (X) into accountable choices (Y) via a decision layer (Z) that prioritizes

reversibility, explainability, and human agency.

0. What This Spec Is, And Is Not

This document defines the formal machinery of a decision-auditing kernel. It states

rules, measurement procedures, canonical sets, formulas, and stop conditions.

It does not contain:

Any scored decision run

Any validation against real-world outcomes

Any claim that the kernel produces correct decisions

The kernel is intended to make a decision's reasoning auditable, not to guarantee the decision's correctness. Two users applying this kernel to the same inputs must arrive at the same output. Whether that output is wise is a separate question.

1. Core Governance Rules

Rule 1 — Weighting (Rank-Reciprocal)

Driver weights are derived from the user's strict ordinal ranking of drivers:

W

$$w_i = (1 / R_i) / \sum$$

$$j (1 / R_j)$$

where R

R_i is the rank of driver i (1 = most important), and the sum runs over all drivers.

No user-assigned weights. No negotiation. Rank-reciprocal is chosen over rank-sum because top-ranked drivers are typically qualitatively more important than lower-ranked ones; linear spacing understates this.

Rule 2 — Driver Orthogonality

Each driver must measure a quantity not already measured by another driver. If a proposed measurement procedure bleeds into an adjacent driver's territory, the procedure is rejected, not reweighted. If the user believes the overlap represents a real preference, the remedy is to re-rank the drivers — not to widen a definition.

Rule 3 — Attested Utility (U_z)

The final score for each option is the Attested Utility:

U

$$U_z = (C_i \cdot U) + ((1 - C$$

$$i \cdot U) + ((1 - C$$

$$i) \cdot R$$

$$_s)$$

where:

U is Intent Utility (the rank-weighted sum of driver scores)

R

—

s is the Reversibility Score

C

—

i is the Certainty Index, derived (not declared)

When certainty is high, U_z tracks intent. When certainty is low, U_z tracks reversibility.

Reversibility is not a primary objective; it is a risk-modulation layer that activates under uncertainty.

Rule 4 — Definition Independence

No metric may be justified by its effect on option scoring. All measurement procedures must stand on their own. A rule against outcome-directed definitions.

Rule 5 — Hard-Filter Resolution

Hard filters (user-declared Non-Negotiables) eliminate options, not decisions. If one option fails a hard filter, it is dropped from the comparison. If all options fail a hard filter, the Abstention Protocol's filter-breach clause triggers. This distinguishes "one bad option" from "the decision is mis-framed."

2. Intent Utility (U)

$$U = \sum$$

$$_i (W_$$

$$i \cdot D$$

$$_i)$$

where D

_i is the normalized [0, 1] score of driver i on the option being evaluated.

Handling current/projected splits (D2, D3):

For drivers with a current/projected split, the spec requires two U values:U

current uses D

—

_xa (current) values where splits exist

U

_projected uses D

_xb (projected) values where splits exist

The user must declare which horizon the decision is being made for. U_

z is then

computed from the U corresponding to that horizon. Silent averaging is forbidden.

3. Driver Taxonomy

D1 — Autonomy / Decision-Sovereignty

Measurement: Veto-to-Permission Ratio. (Decision domains with unilateral veto) /
(Total strategic decision domains in the canonical set).

Canonical Domain Set (fixed for v0.1):

1. Product Roadmap
2. Technology Stack
3. Hiring / Firing
4. Budget / Resource Allocation
5. Strategic Partnerships
6. External Communications / Brand
7. Legal / Policy

For each option, the user marks each domain as Unilateral (1), Shared (0.5), or
Permission-Based (0). D1 is the mean across the seven domains.

D2 — Thesis Validation / Attestation Reach

Measurement: Verifiable external references to the user's thesis (here, the Z-layer), counted over a declared window.

Required Split:

D2a (Current): Attestations existing at the decision moment.

D2b (Projected T+18m): Expected attestations at 18 months given continued execution.

D2a and D2b are separate values. Combining them is forbidden at the driver level; they flow into U
current and U

—

_projected respectively (see §2).D3 — Scale of Impact / Intent-Node Count

Measurement: Number of human or agent decision paths directly governed or constrained by the user's work product, where attribution is traceable to the individual.

The distinction that matters: contribution $\neq$ governance. A team member at a large company contributes to institutional impact; their personal attributable impact is bounded by their decision domain. Reach (impressions, users touched) is not Impact in this sense.

Required Split: D3a (Current) / D3b (Projected T+18m), same treatment as D2.

Scale (logarithmic, [0, 1]):

1.0: $\geq 10^8$ attributable nodes

0.5: $10^4 - 10^6$ nodes

0.1: $< 10^2$ nodes

Linear interpolation on log-count between anchors.

D4 — Wealth / Financial Security

Measurement: Runway Delta. Net change in “months-to-insolvency” at a fixed expenditure baseline, per unit time worked on the option.

Scale:

1.0: Exit / financial independence achieved

0.5: +1 month of runway earned per month worked

0.1: Runway decreases despite income (negative cash flow)

Linear interpolation between anchors.

Financial drivers may only be populated from actual cash-flow data or declared

“Withheld (computed privately).” Estimated values are forbidden.

D5 — Personal Stability

Measurement: Protocol Variance. Fraction of canonical environmental variables that are predictable at a T+30 day horizon.

Canonical Variable Set (fixed for v0.1):

1. Geographic Location2. Sleep / Circadian Window

3. Recurring Daily Obligations

4. Travel / Commute Cadence

5. Meeting / Interruption Density

6. Hardware / Tooling Access

For each option, each variable is marked Predictable (1) or Not Predictable (0). D5 is the mean.

4. Reversibility (R_s)

R

—

$s = 1 - \text{Mean}(E_{f1}, E_{f2}, E_{f3}, E_{f4})$

Each E

_f component is friction (higher = harder to reverse), normalized to [0, 1].

E

f1 — Time-to-Exit

—

Months required to cleanly unwind all contractual and operational obligations.

Normalization: Linear on months. 0.0 at ≤ 7 days. 1.0 at ≥ 24 months. Linear interpolation between.

E

_f2 — Liquidation Penalty

Percentage of total realizable value (IP, equity, cash, accrued benefits) forfeited upon exit. Used directly as a [0, 1] value.

E

_f3 — Identity Lock-in (interpretive)

Persistence of the role-label in the user's public attribution after exit.

Normalization (ordinal anchors, linear interpolation):

0.0: No persistent public association with the entity

0.5: Role appears in professional attribution for ≥ 12 months post-exit

1.0: User's public identity and the entity are indistinguishable

Caveat: This component is inherently more interpretive than E_

f1 and E

_f2. See §7

Caveat 4.E

_f4 — Network Decay (interpretive)

Erosion of external opportunity-set during tenure.

Normalization:

E

—

$f_4 = 1 - (\text{plausible next-step opportunities at exit} / \text{plausible next-step opportunities at entry})$

Clamped to [0, 1]. “Plausible next-step opportunity” means a concrete role, project, or relationship the user could plausibly move to within 90 days.

Caveat: Opportunity counting is interpretive. See §7 Caveat 4.

5. Certainty Index (C_i)

C

$C_i = \text{Mean}(C_{ic}, C_{id}, C_{ir})$

Caveat: Arithmetic mean is a v0.1 choice, not a derivation. Minimum (most honest for a certainty measure) is a candidate for v0.2. See §7 Caveat 1.

C

C_{ic} — Conflict Signal

Measures internal coherence of the top-ranked drivers. Does not reference the user’s actual past choice.

Procedure:

1. Identify the top-2 weighted drivers: $D_{\text{rank}=1}$, $D_{\text{rank}=2}$.
2. For each, identify which option has the higher normalized score.
3. Apply:

Both drivers favor the same option → 1.0

Drivers favor different options → 0.5

Either driver has tied scores across options → that driver contributes 0.5

regardless

Either driver cannot be scored due to insufficient data → 0.0

C

C_{id} — Data Quality (Fix A)

C

$C_{id} = (\text{drivers with current-state observed scores}) / (\text{drivers eligible for current-state scoring})$ Projected-only drivers are excluded from both numerator and denominator. This removes the incentive to hide projection behind a combined score.

C

C_{ir} — Rank Distinction

—

C

—

$C_{ir} = 1 - (\text{number of indistinguishable driver pairs} / \text{total pairs})$

With a strict no-tie ranking, $C_{ir} = 1.0$. When the user reports that two drivers were “a toss-up,” those pairs count as indistinguishable.

6. Abstention Protocol

The Kernel must return “No Verdict” under any of the following:

1. Epistemic Failure: C_{id}

$C_{id} < 0.3$. Output: “Decision conditions insufficient. Gather more information or defer.”

2. Margin Failure: $|U_z \text{ difference between top options}| < 0.05$. Output: “Paths indistinguishable within model margin. Choose on a secondary criterion the kernel

does not model.”

3. Filter Breach (total): All options fail a user-declared hard filter. Output: “Decision is mis-framed. Reframe the option set.”

4. Single-Option Filter Breach: If only some options fail a hard filter, those options are dropped (Rule 5). No abstention; proceed with surviving options.

A kernel that always produces a verdict is a kernel that lies under uncertainty.

Abstention is a feature, not a failure.

7. Open Caveats (v0.1 Limitations)

Caveat 1 — C

C_i aggregation function. C

C_i uses arithmetic mean of its three sub-

—

indices. This may overstate certainty when one component is near zero. Minimum ($C_i = \min(C_{ic}, C_{id}, C_{ir})$) is a defensible alternative and is a candidate for v0.2. Current choice is provisional.

Caveat 2 — C

C_{ic} quantization. C

C_{ic} uses discrete values (1.0 / 0.5 / 0.0), which cannot

distinguish mild from sharp conflicts between top drivers. A continuous version —

scoring conflict by the utility gap between the D_1 -preferred option and the U-preferred option — is a candidate for v0.2.

Caveat 3 — U

U_z integration function. U

U_z uses linear interpolation: $U_z = C_i \cdot U +$

$(1 - C_i) \cdot R_s$. Alternative formulations (multiplicative, penalty-based, bounded-

influence) may produce different rankings. Linear interpolation is chosen for its simplicity and its clean limit behavior (C_{f1}

$$i \rightarrow 1 \Rightarrow U$$

—

$$z \rightarrow U; C_{\text{f2}}$$

$$i \rightarrow 0 \Rightarrow U$$

$$z \rightarrow R$$

—

C_{f3}).

Empirical validation is required before claiming this is the correct form.

Caveat 4 — Interpretive components. E_{f3}

E_{f4} (Identity Lock-in) and E_{f4} (Network

Decay) have measurement procedures that are more interpretive than E_{f1}

and E_{f2} .

—

Two users scoring the same situation may disagree by ± 0.2 on these components

without either being wrong. This introduces real uncertainty into R_{f1}

such that the current

spec does not propagate into C_{f1}

i.

Caveat 5 — No empirical calibration. No driver score thresholds, no E_{f1}

f

normalizations, no abstention thresholds have been validated against real decisions. All

numeric anchors (0.3 certainty floor, 0.05 margin tie, 24-month E_f1 saturation, etc.) are plausible defaults, not measured ones.

Caveat 6 — Self-audit limitation. A kernel applied by the author to the author's own past decision has a credibility ceiling that no amount of internal rigor can raise. First-party backtests should be treated as illustrative, not evidentiary.

8. What a Compliant Execution Looks Like

A kernel run is compliant if, and only if:

1. The user declares rank ordering of D1–D5 before scoring any option.
2. The user declares hard filters before scoring any option.
3. The horizon (current / projected) is declared before computing U.
4. Every driver score is either (a) computed from declared inputs via the stated measurement procedure, (b) marked "Withheld (computed privately)," or (c) marked "Insufficient data."

5. C

$_i$ is computed from the data, not declared by the user.

6. If any abstention condition triggers, the kernel reports "No Verdict" and stops. It does not produce a ranking "anyway, for reference."

7. The full set of inputs, scores, and intermediate values is preserved. A run that cannot be re-derived from its inputs is not a compliant run.

Any execution violating any of the above is not a Z.DecisionKernel run, regardless of how it is labeled.

9. Versioning & Governance

This document is frozen at v0.1.0-alpha as of the release date.

Changes require a version bump and a changelog entry. Silent revision is forbidden.

v0.2 candidates listed in §7 are aspirational, not committed.

First-party backtests against this spec may illustrate the machinery but do not constitute empirical validation.

Attribution line in §Metadata is load-bearing and may not be altered by downstream users.

End of specification.

Declared by:

Rishabh Banerjee

Date : 26 April 2026

Tab 2

